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Energy cost minimization for unrelated parallel machine scheduling under real time and demand charge pricing

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ABSTRACT

There is an increasing concern in manufacturing systems to reduce electricity costs and lower carbon footprints. There are usually two ways of reducing manufacturing electricity costs. One consists of reducing the total electricity demand by using more efficient equipment and processes, and the other focuses on scheduling algorithms to determine the best response to pricing policies put in place by energy companies. In parts of the United States these pricing policies consist of both a consumption charge, which is related to the total amount of energy consumed, and a demand charge for the highest average power consumed over a given time window. Consideration for both time-dependent consumption charges and demand charges when planning a production schedule (e.g. parallel machine scenario) can reduce the total electricity costs. In this paper, a mathematical optimization model is proposed to schedule jobs in which both demand charges and consumption charges are considered for a facility with parallel machines. Results show that total electricity costs can be reduced significantly when demand charges are included in scheduling decisions versus considering only consumption charges. Results further indicate a significant reduction in peak demands for the facility, which provides a benefit for both the facility and the utility. Results also indicate that considering both demand and consumption charges does not increase the total electricity consumed. This study further analyzes the implications of changing model parameters such as the number of machines, machine utilization, and the price of the demand charge. The results provide a better understanding of how demand charges can affect the schedule of a manufacturing facility if total energy costs are to be minimized.

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1. Introduction

Electric utilities face the difficult task of providing an uninterrupted high quality power supply to a set of electricity consumers with highly variable demand patterns. One challenge of particular concern to utilities is meeting energy production requirements during periods of peak demand levels. To meet these infrequent periods of peak demand expensive peaker generation units are employed. System level peak demands occur infrequently, but the cost to meet these peak demands is high, incentivizing utilities to reduce such levels.

One way to reduce system level peak demands is to convince consumers to use less electricity during potential peak periods through a demand response process. There are two main forms of demand response: dispatchable and non-dispatchable (FERC, 2010). With dispatchable demand response utilities may have direct control over a customer's appliances such as HVAC equipment or water heaters, and can disable them if needed. In non-dispatchable demand response customers decide whether to reduce consumption or not in response to signals from the utility. In the United States, two common non-dispatchable demand response approaches for industrial and commercial customers are Time-of-Use (TOU) and Real Time Pricing (RTP) pricing policies. Under TOU the price of electricity changes throughout the day. TOU pricing segments a day into periods of peak, mid-peak, and off-peak hours, and each span of time typically lasts several hours or more. Utilities can increase the cost of electricity during periods of time when system level peaks are likely to occur to incentivize consumers to shift their usage to lower cost time periods. Real Time Pricing (RTP) rate structures are similar to TOU rates, with the exception that the prices change more frequently and at smaller intervals. The dynamic price schedules of RTP are communicated to customers in advance, often with a lead time of 1 day (Nezamoddini

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and Wang, 2017).

TOU and RTP structures encourage customers to shift their load to off peak times, and significant work has been done to develop energy aware scheduling applications for industrial customers, such that they can best respond to changing electricity prices. However, consumption charges, measured in kilowatt hours (kWh), are not the only electricity charges that industrial customers face. Industrial customers also face a demand charge related to their rate of power consumption, measured in kilowatts (kW). For industrial and commercial users, power requirements can vary greatly within a day and across customers. Since electricity cannot be stored and must be generated and supplied to each customer instantaneously, meeting customer requirements for large power rates calls for utilities to have spare and expensive reactive capacity. To account for these high costs utilities charge customers for the availability of large instantaneous power (NationalGrid, 2016), resulting in a separate demand charge for consumed power levels. According to a survey of 43 utilities in the United States that offered TOU pricing structures, 36 of 43 surveyed TOU rate plans include demand charge pricing (Wang and Li, 2015). Similarly, all 5 of the representative utilities offering RTP policies analyzed in Nezamoddini and Wang (2017) had additional demand charges. To date there have been numerous parallel machine scheduling algorithms designed to take into consideration energy consumption charges including Time Of Use (TOU) and Real Time Pricing (RTP) based pricing policies, but very few have also considered the separate demand charge facilities may face.

According to the U.S Department of Energy (U.S. Department of Energy, 2012), utility companies do not use the value of the instantaneous peak load to calculate the demand charge, instead they use a rolling window of fifteen to 60 min to calculate the average power demand over a period of time specified by the rolling interval. The maximum average demand after scanning over an entire month is used to calculate the demand charge (U.S. Department of Energy, 2012). An example depicting the difference between the peak load (kW), maximum average demand, and the total energy consumption (kWh) is shown in Fig. 1. The area under the curve represents the total energy consumption in kWh, while the highest instantaneous demand of 8 kW represents the peak load, and the maximum average demand of 6.8 kW would be used to calculate the additional demand charge.

While TOU and RTP rate structures are designed to reduce the *system* peak load, they may have the opposite effect on individual industrial customers, potentially increasing their individual peak load as they shift production away from high price time periods. An example is presented in Section 5.1 of this phenomenon occurring when demand charges are ignored. This work develops a model to minimize electricity costs in a parallel machine scenario where both real-time pricing and demand charges are considered.

This work specifically considers an unrelated parallel machine manufacturing scenario. In this type of scenario, at the beginning of a production cycle, a known set of N jobs needs to be assigned to run at specific times on a set of M machines. Additionally, the optimal times to transition machines between production states with different energy requirements is also decided. Considering a parallel machine scenario, this study aims to investigate whether the inclusion of premium charges from utility companies in the form of demand charges can potentially offset any gains made in avoiding expensive pricing periods in a real time pricing (RTP) scheme. In this study a real time pricing policy is evaluated where the utility provides the hourly prices of electricity a day ahead of time, so that planning decisions can be made accordingly. Of the 11 RTP programs reviewed in Nezamoddini and Wang (2017), 9 of them offered day-ahead price signals. The premise of this work is that both demand and consumption charges need to be considered when minimizing energy costs, and that considering both demand and consumption charges will not increase the total energy required to meet production goals.

In order to include demand charges, multiple parallel machines, and machine energy states, this research builds upon aspects of two models in the literature. The first model proposed by Shrouf et al. (2014) includes a RTP schedule and machines with different energy states for a single machine without considering demand charges. The second model proposed by Ding et al. (2016), minimizes the time of use based energy costs in a multi-machine parallel schedule, but also does not consider demand charges. The purpose of this study is to understand what happens when both demand and consumption charges are included in the objective of a parallel machine schedule with excess capacity and machines with different energy states. The sensitivity of the developed model is analyzed with respect to different demand charge costs, different number of machines (2, 5), different number of jobs per machine, and system utilization in order to understand the impacts to peak demand, total energy consumed, and total energy costs.

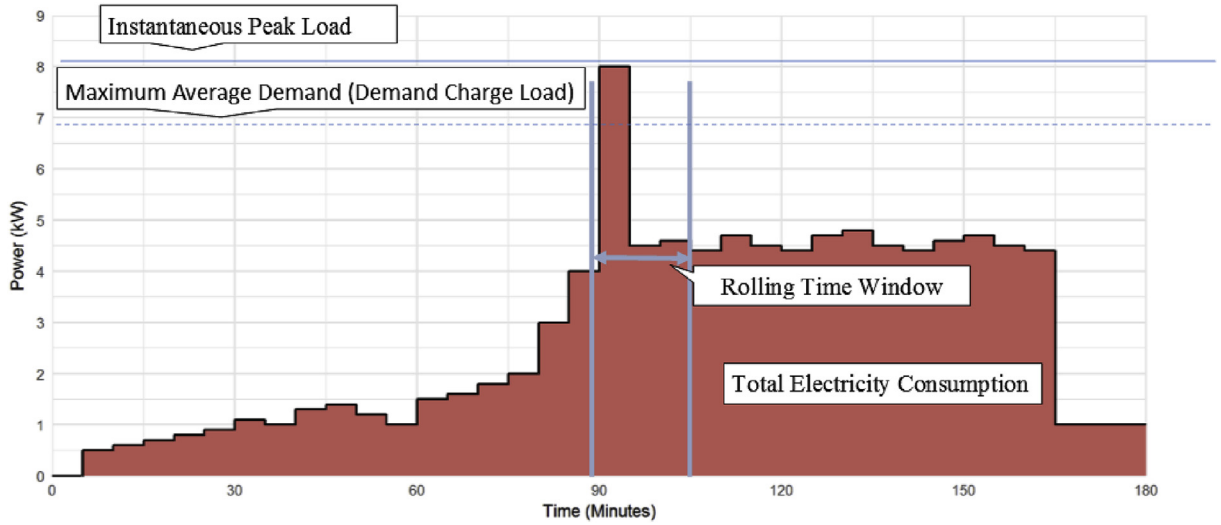


Fig. 1. Peak load versus general demand of electricity.

2. Literature review

This section presents existing literature on methods to schedule parallel machine production systems in order to reduce their electricity consumption, electricity costs, or both. First we review efforts that have focused on reducing electricity consumption, but that have not considered electricity pricing policies. Second we explore efforts to reduce electricity costs when real time pricing and time of use pricing policies are considered. A few special cases in which peak load was considered will also be discussed. Finally, we describe two models that are used as a baseline to the model developed in this paper. While scheduling production with energy related goals has been evaluated for a variety of manufacturing settings, this review will focus primarily on parallel machine environments.

2.1. Scheduling to reduce total electricity consumption

Several avenues have been pursued when scheduling to reduce total energy consumption. One avenue was to recognize that the energy required to complete a job may be machine dependent, where the energy requirement is driven by machine characteristics such as processing speed and/or age of equipment. Li et al. (2016) proposed a model to schedule production in a parallel machine scenario that considered both tardiness and energy costs, where processing times per job determined production energy requirements. Liang et al. (2015) also considered processing time per job as a contributor to total energy consumption. Liu et al. (2014) formulated a model to minimize the weighted tardiness of jobs and the total electricity consumption in a job shop scheduling algorithm. Liu et al.'s work concluded that reducing electricity needs will increase the tardiness of jobs. Job scheduling on parallel machines has similarities to scheduling computation tasks on processing servers. Angel, Bampis and Kacem (2012) developed a model to minimize average weighted completion time and energy consumption for a set of computation tasks. Their model included the consideration that the processing time per job is machine dependent.

Models made to reduce energy consumption often include considerations for machine states. A significant amount of energy is wasted when machines sit idle; therefor many scheduling algorithms targeting energy consumption reduction have included the notion of production or machine states which capture the different amount of energy required for processes such as start-up, idling, production, and shut-down activities. The focus of these models is not only to assign jobs to machines, but to also change machine states at strategic points in time to save energy (such as turning a machine from idle to off). Scheduling models to reduce energy consumption and tardiness for a single machine environment while considering machine states was evaluated by Mouzon and Yildirim (2008). Machines states, and the ability to switch a machine between on, idle, processing and off states were also included in Li et al. (2016) and Liang et al. (2015).

Finally, a third way of minimizing energy consumption was to vary the speed of a machine while processing a job. In models considering processing speed there is a tradeoff between energy consumption and tardiness, where higher processing speeds will ensure more on time jobs while at the same time increasing energy demands. Tang and Dai (2015) proposed an algorithm to minimize total energy consumption by adjusting the cutting speeds of operations while maintaining the original assignments and processing sequence of each job. Using this approach, the authors successfully managed to reduce electricity consumption by 5%–9% while maintaining the same makespan in a small-size instance (less than 10 machine tools). Zhang and Chiong (2016) developed a multi-objective hybrid genetic algorithm to minimize energy consumption and total weighted tardiness in a job shop. Their algorithm considered a finite set of machine speeds, where higher speeds were related to faster processing times and increased energy demands. Finally, Angel et al. (2012) also considered different processing speeds in their model to minimize energy consumption and completion time for computer networks.

2.2. Scheduling to reduce total electricity costs

Another group of energy aware parallel machine scheduling algorithms specifically targets minimizing energy costs. Similar to models that target minimizing energy consumption, models to minimize energy cost also consider machine states (Cheng et al., 2017; Van Den Dooren et al., 2017) and machine or job specific energy requirements (Che et al., 2017; 2017, Cheng et al., 2018). What sets apart most models that consider minimizing electricity costs from models that strictly target minimizing energy consumption is that most models that minimize electricity costs include time of use (TOU) pricing policies in their models.

Moon et al. (2013) introduced a model that was later improved by Koo and Kim (2016) to minimize electricity costs and a weighted overtime penalty cost. The model allowed machines in a parallel machine environment to consume zero energy while idling, and took advantage of this to schedule idling periods during times of peak electricity pricing. Another model that analyzes the TOU tariff is the one proposed by Ding et al. (2016) which minimizes the total electricity cost while keeping the time to complete the jobs under a predetermined production deadline. Ding et al.'s model considers machine and job dependent processing times and power requirements, but does not consider changing the states of machines. Zeng, Che and Wu (2017) developed a bi-objective scheduling algorithm to minimize both energy costs and number of machines utilized. Energy costs considered TOU pricing and were driven by machine processing speeds and job dependent processing requirements. While this model did not explicitly consider demand charges, reducing the number of total machines could have significant impacts to peak demands.

Parallel machine scheduling models to minimize energy costs do not all consider time of use energy prices. For example, Lu and Qiao (2017) evaluated a parallel machine scheduling problem to minimize energy costs for a steel production enterprise where batching was allowed during the heating process. Energy costs were minimized by developing schedules that took into consideration job requirements (heating time and soaking time) that varied per machine. Fang and Lin (2013) considered the ability to change the speed of each job in a parallel machine computing environment. They developed a particle swarm optimization approach to efficiently minimize tardiness and power costs.

While many articles consider time of use energy pricing to minimize energy costs, few if any models considered demand charges simultaneously with consumption charges.

2.3. Scheduling to reduce peak demand

While there has been considerable work related to scheduling in parallel machine environments while taking into consideration energy requirements, the majority of the work has focused only on minimizing energy costs or minimizing total energy consumption. This section explores the few studies that have considered peak demand in scheduling applications.

Bruzzone et al. (2012) considered minimizing peak demand and tardiness in a flow shop scheduling model. The model developed used a two-stage optimization model, where the first stage consisted of minimizing tardiness and makespan and the second stage

minimized peak load. The results show a 60% peak load reduction while maintaining the same total tardiness, but a 24.44% increased makespan. The results suggest that small relaxations in production schedule to better distribute energy loads can potentially result in large reductions of peak demands (Bruzzone et al., 2012). While Bruzzone et al. minimized peak load, the work did not evaluate the effect on total energy consumption or the total electricity bill including both TOU consumption charges and peak demand charges.

Rager et al. (2015) developed a scheduling model for a parallel machine environment whose objective was to level demand. The application was to avoid large simultaneous energy demands generated by several machines requiring high thermal demands from a central steam system. While leveling demand has a similar effect as minimizing peak demand, their model did not take into consideration TOU energy charges nor optimize for total energy costs.

Nattaf et al. (2015) considered a parallel machine environment with a bi-objective model that included a weighted combination of total energy consumption and maximum demand over peak periods. Their model included batching considerations. While demand over peak periods was included, TOU energy costs and peak demand charges were not considered.

Finally, Sharma et al. (2015) considered three different scenarios for a Flexible Flow System model where the speed in a multi-machine set up was changed to achieve different objectives for a TOU tariff. The first scenario minimized the carbon footprint, the second scenario minimized electrical costs, and the final one balanced both economic and ecologic objectives. The first scenario offered the lowest energy consumption per part produced, the second offered the lowest electricity cost per part, and the combined objective offered a balance of both and a lower makespan. Although Sharma et al. (2015) analyzed the effect of both consumption charges and demand charge under a TOU policy, the demand charge calculations include more than one demand charge per day, which is more than what the US Department of energy suggests (U.S. Department of Energy, 2012).

This work adds to the existing literature by considering both demand charges and consumption charges when developing a schedule for a parallel machine environment in order to minimize total electricity costs.

2.4. Baseline models

A model that has many similarities to the one that is presented in this work was introduced by Shrouf et al. (2014), which aims to minimize electricity consumption charges under the assumption of ample flexibility (i.e. where there is enough time to complete all the jobs and allow for down time) and different machine production states. However, Shrouf et al. (2014) only included a single machine setting with no demand charge. Shrouf et al.'s model is expanded to include more machines by adapting the framework from Ding et al. (2016) that minimizes electricity costs in a parallel machine scenario considering consumption charges.

3. Problem definition

This section proposes a mathematical programming model to minimize total electricity costs for a parallel machine setting under real time pricing and demand charge pricing tariffs. The underlying assumptions of this model are as follows: the model considers that there are N independent jobs that must be completed in a single day. Each job has a required processing time, and must be processed on one of M machines. Each machine can process only one job at a time, and no preemption of jobs is allowed. Each machine can be in one of S states, where the energy consumption of the machine is determined by the state. There is assumed to be enough time in the schedule to allow all jobs to be completed and have time for machines to transition between different states. Jobs do not need to be completed in any specific order, yet the problem can be easily modified to include precedence.

Machine states, pictured in Fig. 2, are similar to those presented by Shrouf et al. (2014) where machines change states and their respective electricity consumption. States for each machine include an "Off" state, during which the machine is completely off and no electricity is used; "Turn On," when the machine starts and has the highest rate of electricity consumption; "Idle," when the machine is on and waiting to begin processing; "Processing," when the machine is processing a specific job; and "Turn Off," when the machine has a cooldown time before going into "Off."

The logic of how a machine can transition from one state to another is shown in Fig. 2, in which the main difference from the Shrouf et al. (2014) study is the inclusion of transitions between "Turn On" and "Idle", and between "Idle" and "Turn Off". According to the logic in Fig. 2, if the current state for a machine is "Off," the machine can remain "Off" or transition to the "Turn On" state. If the current state is "Turn On," the machine can remain in the state or transition to the "Processing," or "Idle" states. If the current state is "Idle," the machine can remain "Idle" or transition to "Processing" or to "Turn Off." If the current state is "Processing," the machine may remain processing or transition to the "Idle" or "Turn Off" states. Finally, if the current state is "Turn Off," the machine may remain in the "Turn Off" state or transition to the "Off" state. The model is formulated in such a way that the rules for transitions can be easily changed to represent other configurations of parallel machine production systems even with a different number of states and relationships than those considered in this study.

In the proposed model a single day or production horizon is discretized into a static number of production periods P. Each job has a required processing time captured in periods, and each machine state transition may require a certain number of periods to complete. For example, the Turn On process may take several production periods. Production periods should be small enough to allow the capture of demand charges over the course of the production horizon.

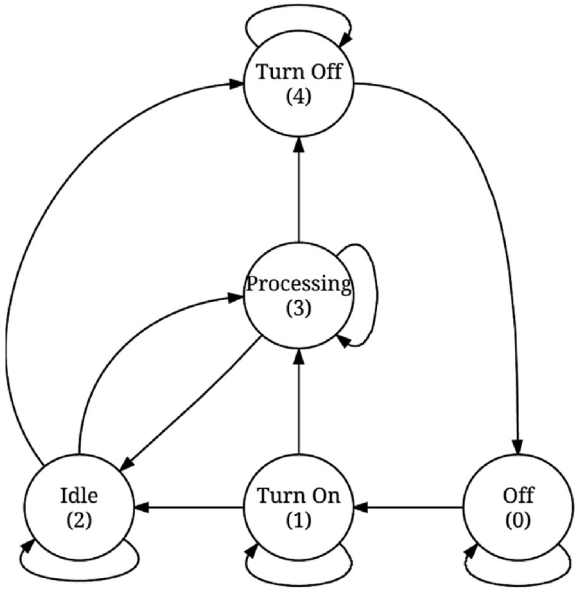


Fig. 2. Logic of allowed state transitions of a machine between periods.

The Parallel Machine Demand and Consumption Charge mathematical model is presented below:

Sets

N : Independent jobs in system

M : Set of machines

P : Periods of the schedule

S : set of possible states of each machine

TS : set of transitional states that require time to be completed (subset of S)

T_s : set of transitions allowed from each state s

Pr : set of states that process parts (subset of S)

Parameters

Pt_j : Processing time of job j

Ep_p : Energy price per k Wp at period p $\frac{\$}{\text{kWp}}$ ($kWp = kilowatt\ period$)

Ec_s : Energy demand (kW) of any machine in state s

Tt_s : Time required to be spent in a transitional state s∈TS

De : Demand charge $\frac{\$}{\text{kW}}$

Rw : Number of periods for peak load rolling window

Variables

w_{isp} { 1, Machine i is in state s during period p; 0, O.W.

x_{ijp} { 1, Machine i is processing job j during period p; 0, O.W.

y_{ijp} { 1, Machine i begins processing job j during period p; 0, O.W.

z_{isp} { 1, Machine i begins transition states s during period p; 0, O.W.

d maximum average demand in a Rw period interval

Parallel Machine Demand and Consumption Charge Model

$$Minimize\ \ Cost = (De{*}d) + \sum_{i\in M}\sum_{s\in S}\sum_{p\in P}\left(w_{isp}Ec_sEp_p\right) \quad (1)$$

subject to:

$$\sum_{j\in N} x_{ijp} = \sum_{s\in Pr} w_{isp},\ \forall\ i\in M,\ p\in P \quad (2)$$

$$\sum_{s\in S} w_{isp} = 1,\ \forall\ i\in M,\ p\in P \quad (3)$$

$$w_{i00} = 1,\ \forall\ i\in M \quad (4)$$

$$w_{isp} \le \sum_{k\in T_s} w_{ikp+1},\ \forall\ i\in M,\ s\in S,\ p\in\{0..|P|-2\} \quad (5)$$

$$\sum_{j\in N} x_{ijp} \le 1,\ \forall\ i\in M,\ p\in P \quad (6)$$

$$\sum_{i\in M}\sum_{p\in P} y_{ijp} = 1,\ \forall\ j\in N \quad (7)$$

$$Pt_j{*}\ y_{ijp} \le \sum_{t=p}^{p+Pt_j-1} x_{ijt},\ \forall\ i\in M,\ j\in N,\ p\in\{0..(|P|-Pt_j)\} \quad (8)$$

$$\frac{1}{Rw}\sum_{i\in M}\sum_{s\in S}\sum_{t=p}^{p+Rw-1} w_{ist}{*}Ec_s \le d,\ \forall\ p\in\{0..\ |P|-Rw\} : Rw\ge 1 \quad (9)$$

$$w_{isp} - w_{isp-1} \le z_{isp},\ \forall\ i\in M,\ s\in TS,\ p\in P : p\ge 1,\ Tt_s\ge 2 \quad (10)$$

$$Tt_s{*}z_{isp} \le \sum_{t=p}^{p+Tt_s-1} w_{ist},\ \forall\ i\in M,\ s\in TS,\ p\in\{1..(|P|-Tt_s)\} : Tt_s \ge 1 \quad (11)$$

$$\sum_{q\in\{0,4\}} w_{iqp} = 1,\ \forall\ i\in M,\ p\in\{(|P|-Tt_4).\ |P|\} \quad (12)$$

The objective (1) minimizes the total cost of electricity, which includes both demand and consumption charges. A total consumption charge results from summing the total energy consumed in each period by each machine multiplied by the energy cost during each time period. A demand charge results from multiplying the maximum average demand of electricity (kW) over a specified rolling time window (calculated by (9)) by the demand charge $\left(\frac{\$}{kW}\right)$.

Each machine state is represented by a number in the model, where 0 refers to “Off,” 1 refers to “Turn On,” 2 refers to “Idle,” 3 refers to “Processing,” and 4 refers to “Turn Off.” Constraint (2) ensures that all machines must be in the processing state when they process any jobs. Constraint (3) limits machines to be in only one state during any time period. Constraint (4) is used to start production after all machines are initially off. Constraint (5) ensures that machine transitions follow only those allowed in Fig. 2.

Constraint (6) ensures that only one job at a time can be processed on a machine. Constraint (7) has two purposes, the first one is to ensure that all jobs get started, and the second one is to ensure that a job can only be started once. Constraint (7) and (8) together ensure that all jobs are finished before the schedule ends and that preemption of jobs is not allowed to happen. Constraint (9)

calculates the maximum average demand in the specified rolling window. Note that for constraint (9) to work the rolling window must be a multiple of the period length. Constraints (10) and (11), along with the variable z, are only needed when the machine has transitional states that require a specific amount of time for such transitions. In the case of the logic shown in Fig. 2, those states will be "Turn On" and "Turn Off." Constraint (10) is used to track the first period of each of the transitional states that require a specific amount of time. Constraint (11) ensures that the transitional state lasts the required number of periods. For example if the "Turn On" process is started in period 3 and takes 2 periods to complete, constraint (11) ensures that the machine will remain in the "Turn On" state for periods 3 and 4, ensuring that the machine finishes the turn on process before it can be assigned to a processing task. Constraint (12) ensures that the machine is "Off" at the period right after the schedule ends, making sure to account for the time required to complete the "Turn Off" state.

4. Methodology and experimentation

A study was designed to understand the impacts of including demand charges in the objective function when minimizing total electricity costs in a parallel machine scenario. The response variables of interest included the maximum average demand over a 15-min rolling window (kW), the total electricity consumed across all machines (kWh), and the total electricity cost ($). Our hypothesis was that systems with greater flexibility, would show greater benefits (i.e., lower peak demands and lower total electricity costs) when demand charges were included in the objective function. However, there was a concern that including demand charges could increase the total electricity consumed when flexibility was high, as machines may spend significantly more time idling. Increasing total electricity consumed would have a negative effect on the carbon footprint of a production facility, as such this potential impact warranted investigation. To explore system flexibility's effect on the response variables an experiment was conducted to evaluate the impact of the number of machines, the machine utilization, and the number of jobs per machine. Demand charge price was also included as a factor in the experiment. More extreme demand charges have the potential to amplify effects of including demand charges in the objective function. The levels tested for each factor are specified in Table 1. Each factor is discussed in more detail in Section 4.1.

Ten replications were completed for each of the 36 factor combinations, resulting in a total of 360 model runs. For each replication, a unique set of randomly generated jobs were created. In order to maintain desired system characteristics such as utilization per machine and number of machines while varying jobs per machine, the processing times of jobs were manipulated. To assign the processing times for each replication, each machine's required utilization time was divided up into parts based on the number of jobs per machine using a multinomial distribution. This ensured that generated schedules were feasible. To create the different replications the only factors needed were jobs per machine, utilization, and number of machines, which with 10 replications resulted in 120 different job lists. A job list consists of a set of jobs that must be completed in a single day and their required processing time. These jobs lists were consistent across all different demand charge prices considered. Each job list should be considered independent of one another. Additionally, the amount of processing time required for a job was considered unrelated to machine power requirements. All jobs used the same power rate while a machine was processing them. For example, if job A required 3 min and job B required 6 min, both jobs would use the same power rate while any machine was processing them, resulting in job B using twice as much energy as job A.

All experimental runs used the same state transitions, required transition times, power required per state of machine, and real-time pricing schedule. Table 2 provides the machine state power requirements and transition time requirements. Table 3 provides the hourly Real Time Pricing (RTP) schedule used. This RTP schedule was created by obtaining sample values of the schedule from the month of July 2016 from Ameren Illinois (Ameren Illinois, 2017). Each run encompassed a single 8-h shift that was divided into 32 equal periods of 15 min. A rolling window of 1 period was used to calculate maximum average demand.

The model was written in AMPL (AMPL, 2017) and solved using the CPLEX Solver (IBM, 2017) running on a Linux server with a 43 core CPU with 256 GB in RAM.

4.1. Factors

The factors of interest of this study are the demand charge prices, number of machines, machine utilization, and the number of jobs per machine. These factors are described in the following subsections.

4.1.1. Demand charge

The demand charge factor is included to understand the effect of demand charge price on the schedules provided by the model. Utility companies price the demand charge differently and the Energy Information Administration does not collect or regulate this price (U.S. Energy Information Administration, 2017). Of the 5 utilities survey by Nezamoddini and Wang (2017) the demand charge ranged from 3.48 $/kW to 18.62 $/kW. The three levels chosen for demand charge (0, 10.2, and 15.3 $/kW) were estimated

Table 1
Factors of the experiment.

Factors	Levels	Detail
Number of machines	2	2, 5
Utilization %	3	70, 80, 90
Jobs per machine	2	2, 4
Demand Charge Price $\frac{\$}{kW}$	3	0, 10.2, 15.3

Table 2
Power and time requirement of each state.

State	Power Required (kW)	Time Required (in periods)
"Off"	0	N/A
"Turn On"	5	1
"Idle"	2	N/A
"On"	4	N/A
"Turn Off"	1	1

Table 3
RTP schedule.

Period	Price US $\frac{¢}{kWh}$
8:00 a.m.	2.7
9:00 a.m.	2.7
10:00 a.m.	2.9
11:00 a.m.	4
12:00 p.m.	4
1:00 p.m.	5.1
2:00 p.m.	4.1
3:00 p.m.	4.2
4:00 p.m.	4

corresponding to 0, 200 and 300 times the maximum hourly rate of 5.1 $\frac{¢}{kWh}$ of the schedule presented in Table 3.

4.1.2. Number of machines

Since demand charges do not affect the schedule of a scenario in which only one machine is included, it is important to know the effect of including the demand charge price when more than one machine needs to be scheduled. To understand the relationship between total energy costs and number of machines two number of machines levels are considered. The first level includes two machines and the second five machines. As more machines are added to the model, more jobs are added such that each machine in the model can meet the required utilization.

4.1.3. Utilization

One of the key assumptions of this study is that there is some flexibility in the schedule to allow for the reduction of electricity costs through efficient job scheduling. Productivity was assessed by considering the utilization rate of the machines in the system. Three levels of utilization were used to mimic utilization rates of 70, 80 and 90%, which over a 32-period schedule, correspond to a machine being used for processing tasks for 22, 26, and 29 periods respectively.

4.1.4. Jobs per machine

Before scheduling production, the model requires the number of jobs to be completed and how much time each job will take. To create the job lists, two levels for the number of jobs per machine were considered: 2 and 4. There are some limitations in the generation of job lists including: the processing times produced need to be integers (discrete distribution), machine utilization needs to equal the levels specified by the experimental designs, and the set of jobs must allow for a feasible schedule. A multinomial distribution was used to generate job processing times by dividing a specific number of periods (utilization) into *n* jobs per machine. As the number of jobs per machine increased, the mean processing time per job decreased. The mean processing time of all jobs for each utilization and jobs per machine combination can be seen in Table 4.

Table 4
Mean processing time of jobs depending on the scenario.

Jobs Per Machine	Utilization	Mean Processing Time (periods)
2	70	11
	80	13
	90	14.5
4	70	5.5
	80	6.5
	90	7.25

5. Results and discussion

This section summarizes results and discusses the influence of the factors of interest on the response variables defined in Section 4. The number of jobs per machine is not included in the following discussion as no significant difference was found between the two job levels tested (2,4). Fig. 3 highlights the similar distributions of the three response variables (average peak load, total electricity consumption, and total energy costs) versus the number of jobs per machine and demand charge price. Therefore, this factor is disregarded and not included; in remaining sections the average response of both 2 job and 4 job per machine levels are reported.

5.1. Maximum average peak load

Fig. 4 shows that the maximum average peak load decreases when the demand charge is considered. Additionally, the peak load remains constant for demand charge between 10.2 $\frac{\$}{kW}$ and 15.3 $\frac{\$}{kW}$. When no demand charge price is considered, the optimization problem only focuses on avoiding the high prices due to the RTP electricity schedule, creating large peaks at the beginning of the day.

Fig. 5 provides an example of the differences in schedules produced both with and without demand charge considerations. Fig. 5(A) represents the energy cost throughout the production horizon. Fig. 5(B) illustrates an optimal schedule to minimize cost when demand charge prices are ignored (set to 0 $\frac{\$}{kW}$). Notice how the schedule presented in Fig. 5(B) causes large peaks in demand during the lower cost periods illustrated in Fig. 5(A) and (C) provides an alternative schedule for the same jobs that minimizes total electricity costs when a demand charge of \$10.2 $\frac{\$}{kW}$ is considered. When considering demand charges the schedule that minimizes costs uses more electricity during the high-priced periods, but achieves savings in peak demand charges.

Fig. 4 also demonstrates that as the number of machines increases, peak load increases as well. However, the same cannot be said about increasing utilization. For the two-machine runs there was no effect on peak load by increasing utilization. For the five-machine setting, when there is high utilization (80% and 90%) the same 25% drop in peak load reduction can be achieved as in the two-machine scenarios. However, when utilization is low (70%), the average peak load reduction is 56.25%. This suggests that the optimization model provides larger benefits when more machines can be used and there is flexibility in the production schedule.

While a minimum of 25% peak load savings can be achieved by incorporating demand charges of at least \$10.20 per kW into an

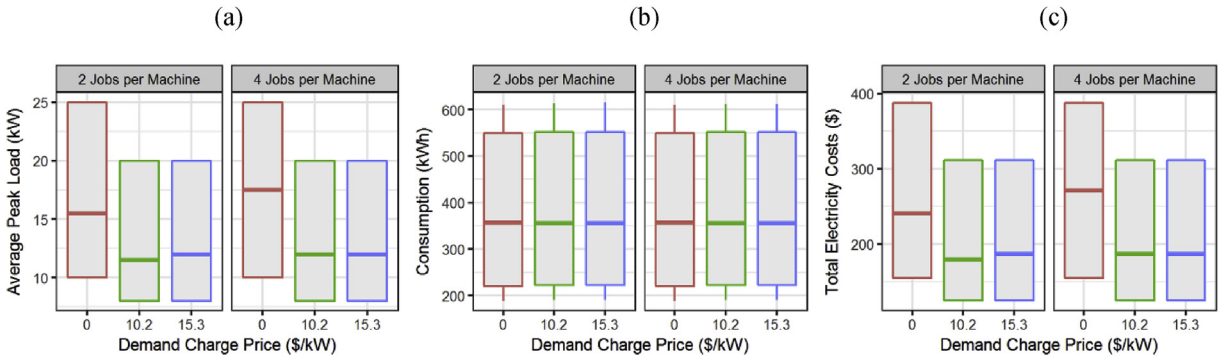


Fig. 3. Comparison of 2 jobs per machine versus 4 jobs per machine for (a) average peak load (b) total electricity consumption (c) total electricity costs.

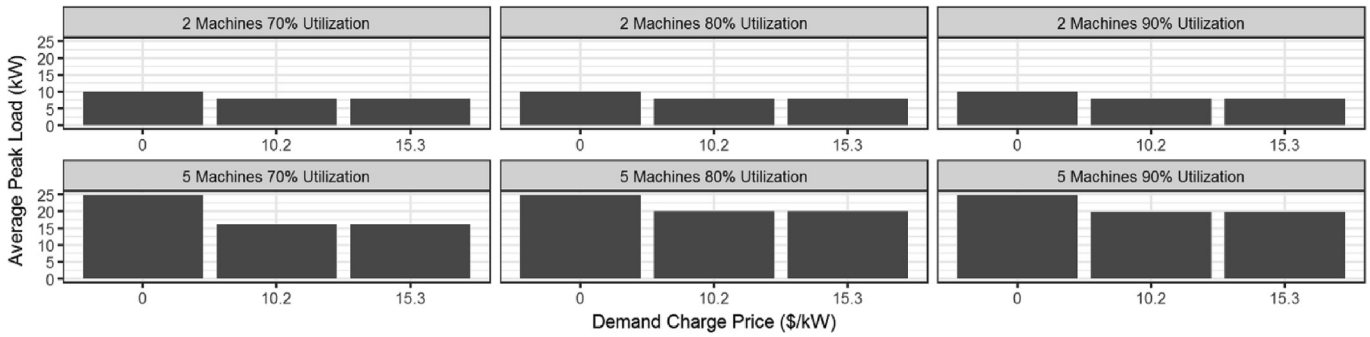


Fig. 4. Average peak load under RTP policy by number of machines and machine utilization.

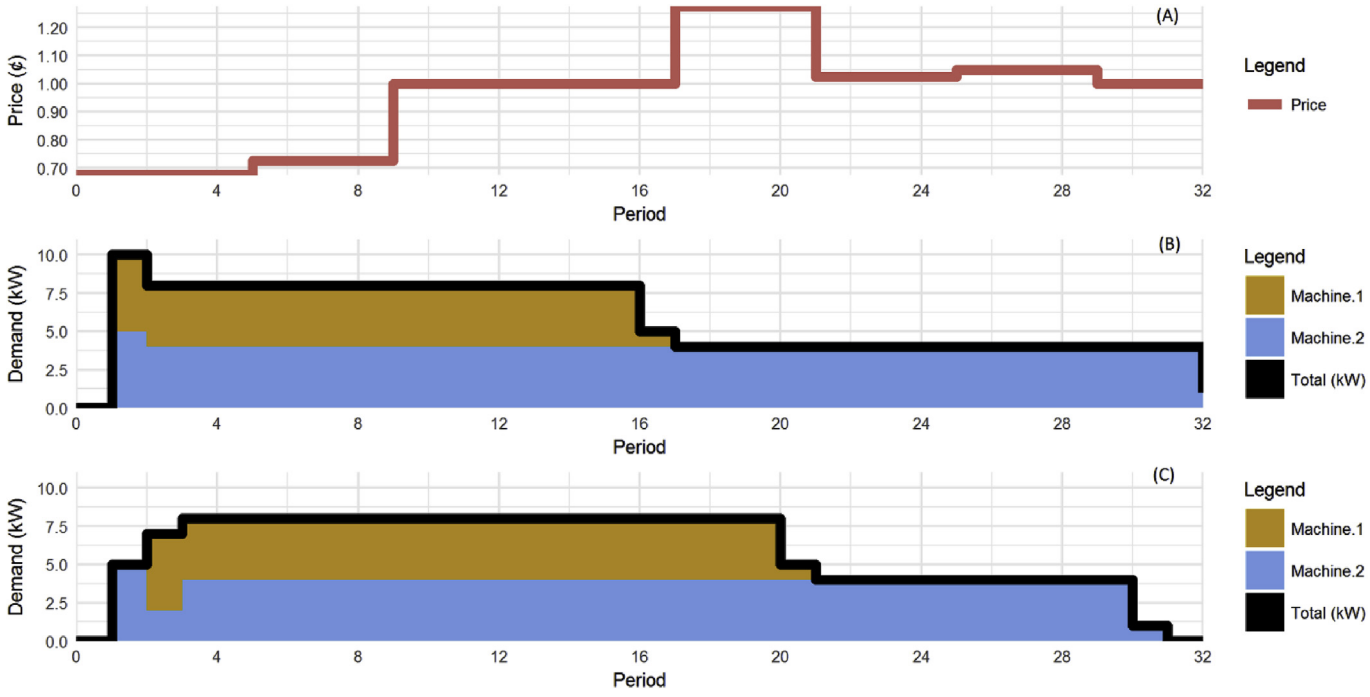


Fig. 5. The energy use of two machines running at 70% utilization with four jobs per machine given the Real Time Electricity prices seen in (A). (B) Energy use of the optimal schedule considering only RTP, and (C) Energy use of the optimal schedule considering both RTP and a demand charge price of $10.2/kW.

electricity cost minimization algorithm, a natural question arises as to what level of demand charge is required before a peak load savings is achieved? To investigate this question different levels of demand charges were explored in increments of $0.20 to observe at which level a decrease in peak load begins to happen.

Fig. 6 shows the maximum average peak load of the 5-machines-70%-utilization scenario as demand charges are increased. Even when electricity companies charge a low rate of 0.2 $\frac{\$}{kW}$, the peak load falls to 16 kW from a starting point of 25 kW. This suggests that even when demand charge prices are low, including them in the model can reduce the peak load significantly.

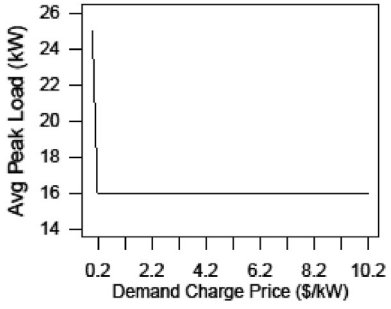


Fig. 6. Maximum Average Peak load as demand charge price increases for the 5 machines, 70% utilization scenario.

5.2. Energy consumption

Fig. 7 shows the relationship between energy consumption and demand charge price, number of machines and utilization rate. The total energy consumption is affected by the number of machines and utilization rates, but demand charge prices have minimal impact to total energy consumption. As utilization and number of machines increase one would expect the total energy consumption to increase because more jobs are used in such scenarios. However, results suggest that as demand charges increase the total consumption largely stays the same. This is an important observation that suggests that a model considering demand charges may not increase total energy consumption, and therefor may not increase

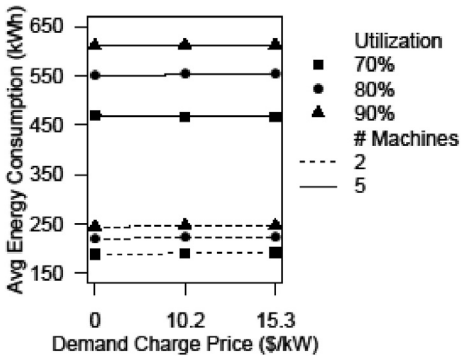


Fig. 7. Average total energy consumption (kWh) under RTP policy with demand charge consideration.

the carbon footprint of a manufacturer in a significant way.

5.3. Total electricity costs

The total electricity cost aggregates the consumption and the demand charges. To understand the tradeoff in total electricity costs between the model proposed in this paper and a similar model that only considers consumption charges we compare only two scenarios, one where the schedule is made by setting demand charges to $0.00 per kW, and one where the schedule is made considering demand charges are set to $15.30 per kW. When demand charges are set to $0.00 per kW the model is equivalent to a model that only minimizes consumption charges. Fig. 8 illustrates that when demand charge prices are ignored in the scheduling process, the total electricity costs are higher than in the other scenarios. However, once demand charges are considered, the cost of electricity is significantly reduced.

The total electricity cost decreases as the demand charge price is taken into consideration in the scheduling process. In the two-machine scenario, the total electricity cost decreased by at least 19% independent of the utilization of the machines. The five-machine scenario achieved similar results at reducing cost at the higher utilization rates (80% and 90%), but the decrease at a low rate of utilization (70%) was of over 34%. Scenarios with more machines and lower utilization rates show greater cost reductions when demand charge prices are included in the scheduling process.

6. Scalability of the model

The model developed in this study used as a baseline the models developed by Shrouf et al. (2014) and Ding et al. (2016) which suggest that this instance of parallel machine problem is NP-hard. The solution time versus the different complexity factors of utilization, number of jobs per machine, and number of machines were recorded and are displayed in Fig. 9. The solution times and variability of solution times increases as the number of jobs per machine increases. This is because the problem expands as more constraints are created when more jobs are introduced. As soon as more machines are introduced, time to solve the model also increases; at 70% utilization going from 2 machines with 4 jobs each to 5 machines with 4 jobs each caused a 40X increase in mean solution time. Utilization shows an interaction effect with the number of machines. When the number of machines is low, as utilization increases the average solution time decreases. However, when the number of machines is higher the highest average solution times were observed with 80% utilization, while 70% and 90% utilization provided faster solution times.

Considering that the factors that contributed the most to longer solve times were increasing the number of machines and the jobs per machine, two different experiments were run to explore the overall scalability of the problem as each of those factors was changed. Fig. 10 shows that the model provides quick solutions for up to 13 machines, but a drastic increase in solution times is observed between 15 machines and 16 machines, and after this point the problem will takes longer than a day to complete. Fig. 11 displays the results of increasing the number of orders per machine from 1 to 16. Interestingly, the highest solve times occur not at the highest number of jobs per machine, but rather in the middle, at 9 jobs per machine. One factor that is affecting this behavior is that even though more constraints are in the problem with high orders per machine, they might be creating a feasible area that is extremely small and therefore decreasing solution times.

Overall this model is able to handle up to 13 machines with 4 jobs per machine, or 5 machines with up to 15 jobs per machine with reasonable solution times without the use of specially designed metaheuristics. Developing a scalable metaheuristic to solve larger size problems has been left for future work.

7. Conclusions and future work

This study evaluated the effects on peak demands, total electricity costs, and total electricity consumed when including demand charges in a parallel machine scheduling mathematical model. Results indicated that considering demand charges in production planning can help to reduce peak load and total energy costs without increasing total energy consumption. The most significant cost and peak load savings are achieved when the number

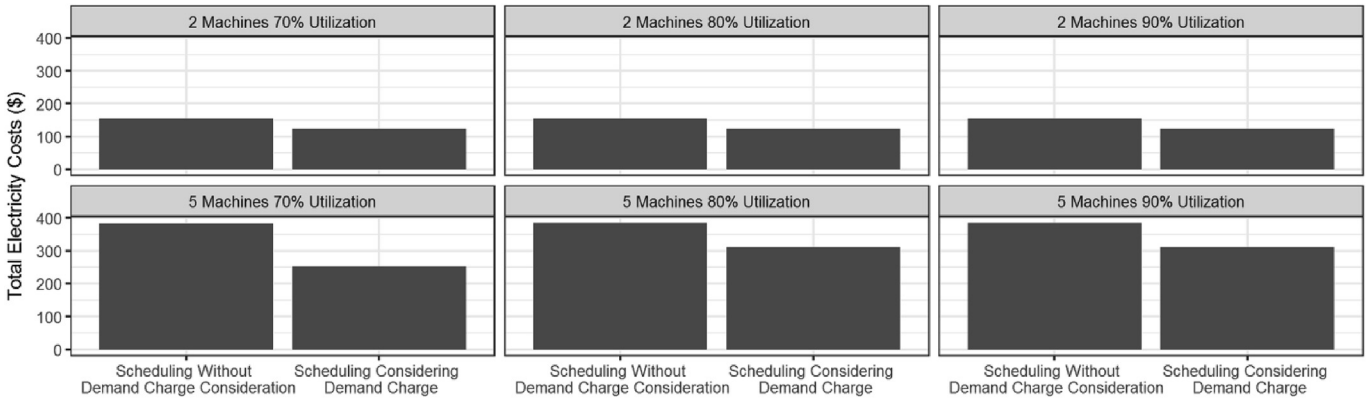


Fig. 8. Total electricity costs under RTP policy and Demand Charges of $15.30/kW.

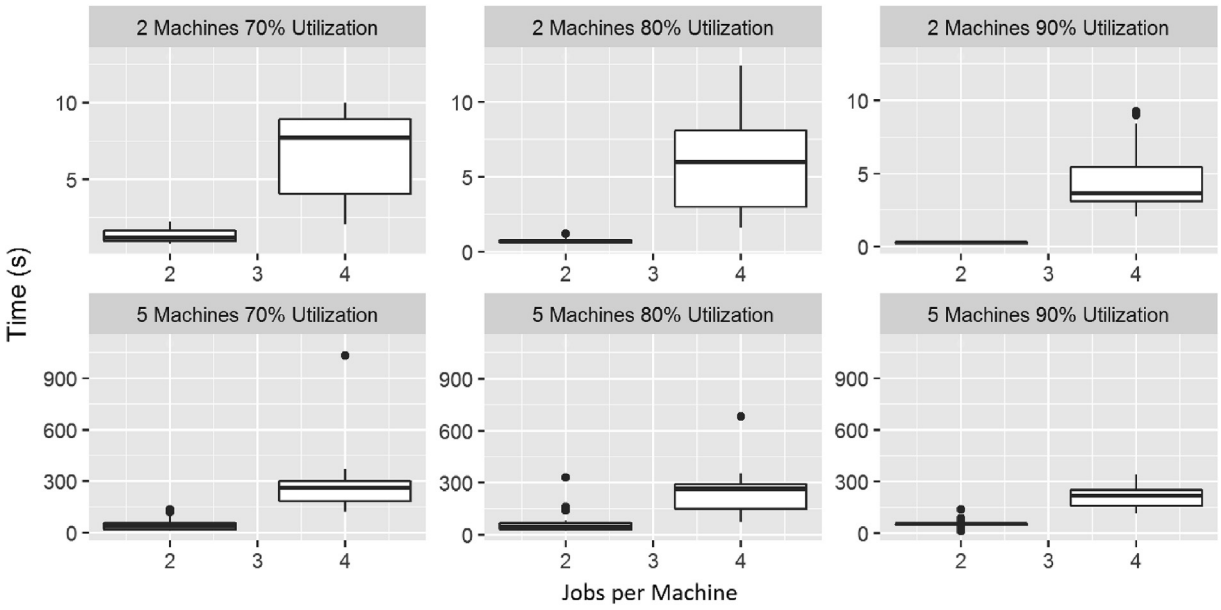


Fig. 9. Solution time versus model factors.

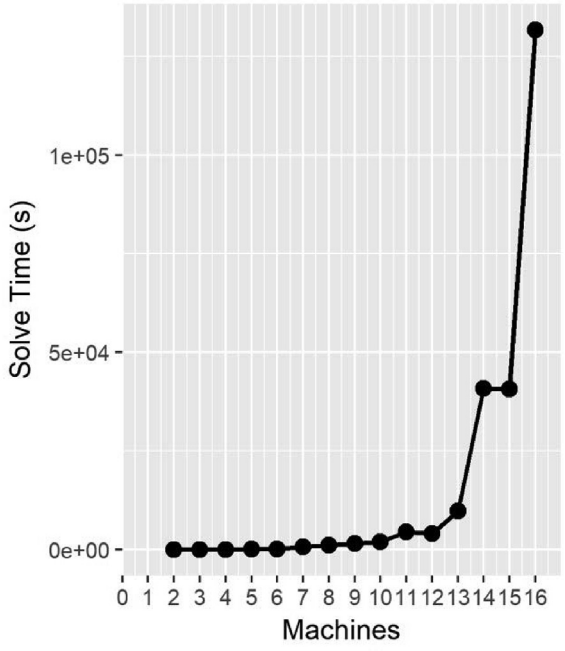


Fig. 10. Solution time of 4 orders per machine running at 80% utilization as number of machines increases.

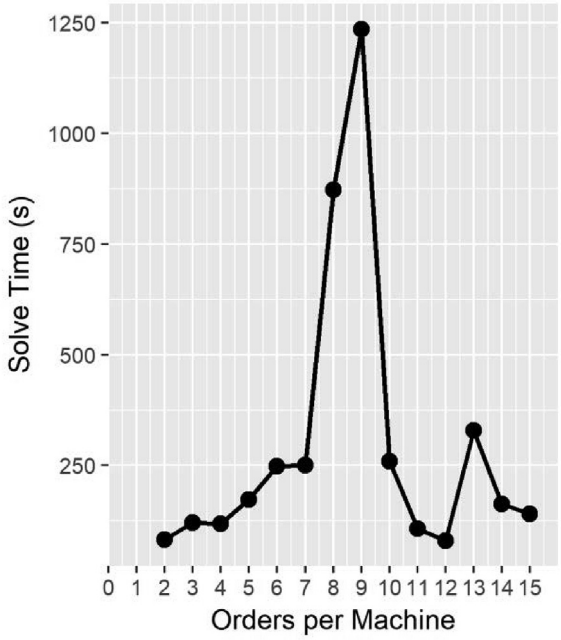


Fig. 11. Solution time of 5 machines running at 80% utilization as orders per machine increases.

of machines are increased and utilization is lowered. Even when the price for the demand charge is as low as $0.20/kW, the schedule avoids high peak loads. Total energy consumption was not significantly impacted by including demand charges, indicating that considering both demand and consumption charges in production scheduling may not increase a manufacturer's carbon footprint. When demand charges exist and are considered in production planning, there is a sharp decrease in total electricity costs.

The results of this study can be used as a benchmark to understand how demand charge prices affect electricity costs when scheduling production for a set of machines. The formulation is flexible enough to allow machines with different states than the ones included in this model. Including demand charges in scheduling applications is a win-win-win situation for customers, utilities and the environment. This is because customers can save money on their utility bills, while utilities benefit from the reduced peak demands. The inclusion of demand charges did not significantly increase the total energy required to complete all jobs which consequently allows a production facility to reduce their costs without increasing their carbon footprint.

A limitation of this study is that the planning horizon was only a single day while peak demand charges are assigned based on an entire month of energy consumption. Therefor our model, when

used on a planning horizon shorter than the billing period, will not guarantee minimal electricity costs. However, this work demonstrated that when using the proposed model, peak demands over the small planning horizon are lowered significantly, and these reductions occur even when demand charge prices are as low as $0.20/kW. Using this observation, one can assume that if the model were run daily to produce production schedules, peak demands would be lower at the end of the billing period than if demand charges were not considered at all. Additionally, a weight could be added to the consumption charge in the model objective to scale a single day's consumption charge to be closer to an estimate of the monthly consumption charges.

Some assumptions were made in this study that need further consideration. This paper provided results of only a single set of day ahead Real Time Prices. Additional studies could be completed to vary these prices, perform a sensitivity analysis of RTP policies, or look at horizons longer than a single day. Furthermore, a model more stochastic in nature could be developed that evaluates the impact of including demand charges when real time prices are not known in advance, but rather change in real time throughout the day.

One of the most important assumptions of the model relies on schedule flexibility, wherein as long as jobs are completed by the end of the day, the schedule will be feasible. This allowed the experiment to concentrate mostly on electricity instead of production requirements, which is in line with what other studies have focused on. A natural extension of this study must jointly consider the electricity costs and production requirements in the face of time of use consumption and demand charges. An interesting addition to the model is introducing multiple objective functions in which both electricity and tardiness of jobs are considered.

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